Jackson Smith

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Education

Purdue University West Lafayette, IN
Bachelor of Science in Computer Science, Bachelor of Mathematics 2028
GPA: 3.21

Experience

Golf Course Maintenance 2827 US-52, West Lafayette, IN
Coyote Crossing Golf Club June 2024 • September 2024

- Operated independently and in team settings
- Managed a large range of equipment and machines
- Applied problem solving skills to manage obstacles
- Worked diligently and consistently under harsh conditions

University Projects

Nupiz

- **Description:** A high level programming language that is self hosted and compiles to a custom bytecode. Designed to be easily accessible and quick to pick up yet allowing more complicated design patterns at request. Fit with static typing and compile time optimizations.
- **Role:** Language Designer, Lead Developer
- **Technologies Used:** C
- **Outcome:** This project has taught me incredible detail about design patterns, code organization, and how lower level structures such as memory and assembly work. It's assisted me in writing code in other languages; understanding how all the systems underneath work. It also challenged me in my ability to research, implement, and make my own creative solution to tricky problems.

Iris (TartanHacks 2026)

- **Description:** Iris is an AI-powered voice-activated Chrome extension that syncs with Google Workspace to reduce friction amongst people with disabilities using Google products.
- **Role:** Frontend Developer, Backend Developer
- **Technologies Used:** Python • AWS Lambda • GPT-4o Mini • OpenAI Whisper • CSS • JavaScript • HTML
- **Outcome:** I built this project in a team of three in twenty-four hours during Carnegie Mellon's annual *TartanHacks Hackathon*. This was an incredibly exciting experience, as we designed an extremely helpful and applicative product for the real world, compared to most gimmicky products. It gained me a lot of experience with OpenAI and LLM systems, and taught me a lot about designing a functioning product with a full tech stack and collaborative team.

DASM2

- **Description:** A fully functional assembly language, compiling to a base 10 machine code system designed for a custom CPU written entirely inside a single compound mathematical instruction inside of Desmos graphing calculator.
- **Role:** Lead Developer, Language Designer
- **Technologies Used:** Desmos • Python
- **Outcome:** This project was an incredible challenge to work on, and pushed me to use my computer architecture knowledge in completeness. In this project I implemented an entire virtual CPU contained within a Desmos graphing calculator project. Not only is it Turing complete, and supports user input and RGB color output, it also has a complete assembly language that compiles using Python. It is typed check, supports stack and register operations, labels, branching conditionals, functions, and anything else needed. The project is lightweight and can run in any web browser. While it is not extremely efficient, it was an amazing project to design and push myself to complete under such harsh constraints.

NeuroEv-AugTopologies

- **Description:** A custom implementation of the NEAT algorithm for machine learning, implemented in Python and trains agents to survive organically through reproduction, plant consumption, and passing on optimal genes.
- **Role:** Lead Developer
- **Technologies Used:** Python
- **Outcome:** This project was super interesting to design, as it is an implementation of the NeuroEvolution of Augmented Toplogies (NEAT) algorithm I read about in an MIT Journal. It uses unique methods such as gene crossovers and a self-building neuron structure, rather than simply adjusting weights and biases, to train agents to survive in a harsh, yet simple, 2D world. It taught me a lot about alternative and more biologically focused machine-learning methods, and their implementations.

Additional

Language Skills: Spanish
Technical Skills: C • OpenAI • Java • Python • HTML • CSS • TypeScript • SQL • SSH • Mongo
Volunteer Experience: Local Unified Sports
Interests/Hobbies: Powerlifting • Game Design • Animation • Math • Personal Finance